UPANDE PACKHOUSE · WORKFLOW

Allocation Planning

Plan the delivery day before allocation starts: who gets which flowers, from which farm, packed by which team, with the stock held so nobody else can take it.

What we built, how it works, and what is left to do
Written 29 October 2026

3	4	4	75
NEW RECORDS	KINDS OF STOCK	HEADS-UPS	TESTS PASSING

WHY

Orders were shorted while stock was on the shelf

We read three weeks of kaitet-group (orders due 7 to 27 October: 1,983 orders, 7,813 lines, 4.55M stems) against pick lists, pack lists, the shelving log, discards and the production forecast. Allocation was happening line by line, against a shelf count that was often wrong, at 8 to 11am the day before delivery.

UNFILLED, STOCK THERE

1.02M

of 2.52M unfilled stems sat where the shelf covered all demand

AVOIDABLE DISCARDS

253k

of 1.08M discarded stems could have filled an order

DOWNGRADES THAT COST

37k

stems went longer while a buyer for that length stayed unfilled

USED CONFIRMATION

0.4%

of lines (31 of 8,243) had a confirmation at all

What we found

- The loss happens at the **variety and length** for a day, but allocation worked **line by line**.
- A confirmation was only a promise: it never reduced what another customer's allocator saw as available.
- 40% of delivered orders (797) never got a pick list. Many were probably offline, which the shelf cannot see.
- 37 mixed boxes and bunches were left half allocated: stock tied up that could not be packed.
- The revised forecast is within about 1% of actual in total, but about 30% off for a single variety in a week.

What we set out to change

- Let sales **plan the whole day** before allocation, with the stock held for the customer it was planned for.
- Plan against **real stock and expected stock**, and be honest about how far to trust a forecast.
- **Warn**, in plain words, when a choice would short another customer or waste stock.
- Make the link to allocation **cheap to check**: allocation reads the holds and releases them.
- Capture the **team** for every line now, so allocation can autofill it later.
- Keep the page the **cleanest in the dashboard**: one list, one action.

Process view. The cause is the process, not the people. Allocation is first come, first served on a live count; the unit of work (a line) is not the unit of loss (a variety and length); and nobody owned the balance of the whole book. Planning gives that balance an owner and a place to work.

HOW IT WORKS

Plan, hold, allocate, release

A planner confirms stems for an order line. Stock on a shelf, or received and waiting for a shelf, is held bucket by bucket for that customer. Allocation cannot give those buckets to anyone else, uses them when it allocates the line, and lets go of the rest.

1 Open the day
Pick the sales farm and the delivery date (tomorrow by default).

2 Confirm
Per line: stems, the farm to take stock from, and the team.

3 Hold
Buckets (oldest first) and stems are held for that line.

4 Allocate
Allocation hides held buckets from other lines and offers them to the line they are held for.

5 Release
Allocating uses the hold; a full line frees the rest.

Four kinds of stock to confirm against, most certain first

On the shelf — held bucket by bucket
free, not too old, not cooling, not held for another line. Shown farm by farm.

Received, not shelved — held bucket by bucket
buckets waiting for a shelf, within the discard age, with their cold room.

Harvested, not received yet — stems only
the cutting day's production on its way.

Still to be cut — stems only
the revised forecast for the cutting day, less what was already harvested, shown by farm.

Stock that does not exist yet is confirmed too, but it is held only when the flowers arrive: the system converts it to real buckets, oldest plans first.

Two farms, not one

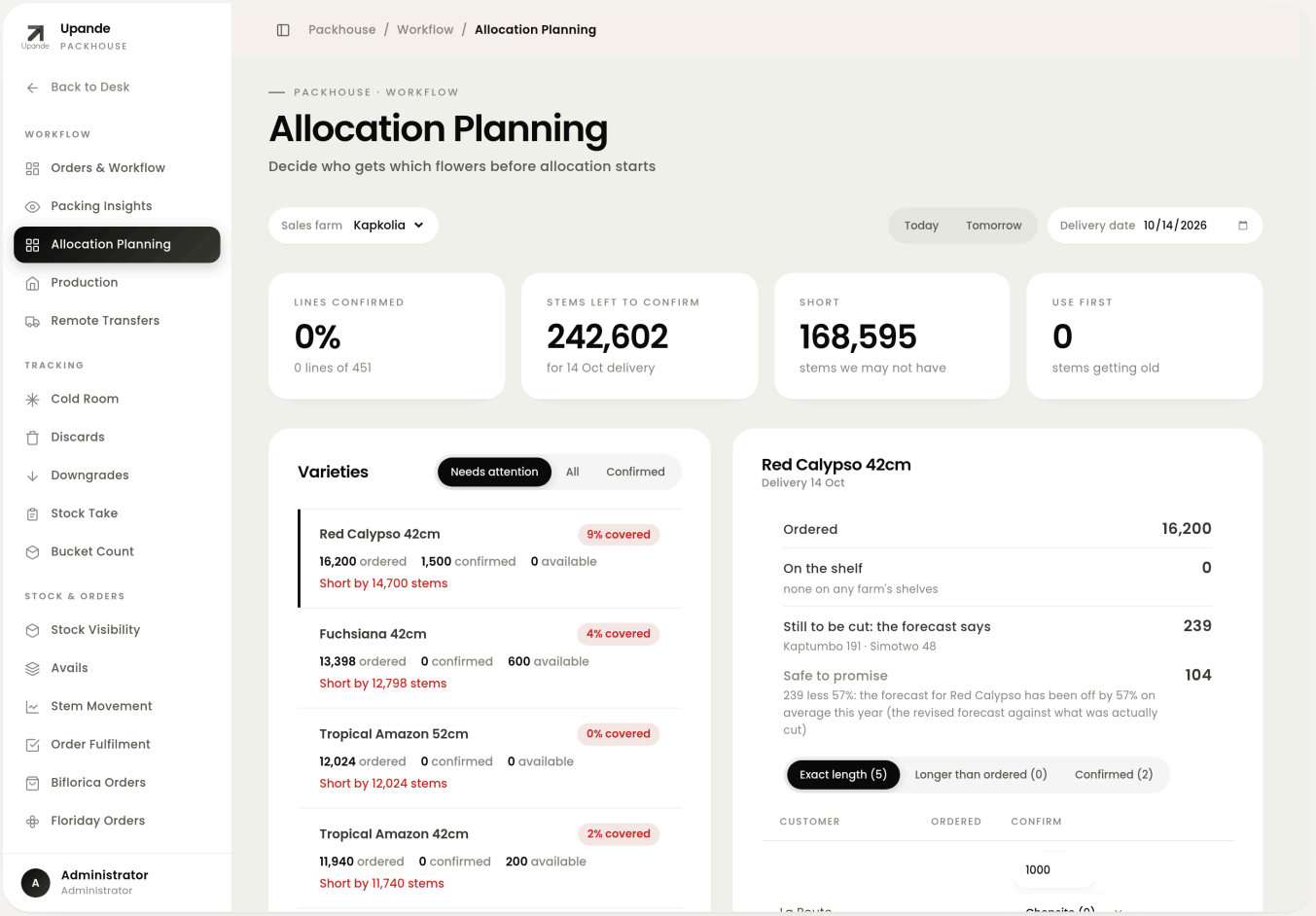
- **S** **Sales farm** is the farm that processes the order. The filter at the top lists only the farms marked as a sales shelf in Production Settings (Kapkolia, Karen).
- **F** **Stock farm** is the farm whose stock covers the stems. Every farm in the sales farm's location is visible, and each customer row has a **From** dropdown.
- **T** **Team** belongs to the order line. A mixed box is packed by one team, so a team chosen on one of its lines applies to the whole box.
- **D** **Delivery date** opens on tomorrow, set the same way as on every v2 page.

What "safe to promise" means

The forecast is not always right, so for each variety we compare the revised forecast with what was actually cut this year and take that average error off. If Red Calypso has been off by 57% on average, 239 stems still to be cut counts as **104 safe to promise**. The page shows both numbers and the sum between them. A variety with no finished week to judge by counts none of its forecast as safe.

THE PAGE

One list, one detail, one action



Allocation Planning on a demo day (14 October) on the development site. The stock shown is old, so most varieties read as short.

1. **Filter bar.** Sales farm, Today / Tomorrow, and the Delivery date.
2. **Four numbers.** Lines confirmed, stems left to confirm, short, and use first (stock that is getting old).
3. **Varieties.** Most urgent first, in three views: Needs attention, All, Confirmed. Each row says how covered it is and why.
4. **Detail.** What exists for the variety and length, then the customers, in three tabs.

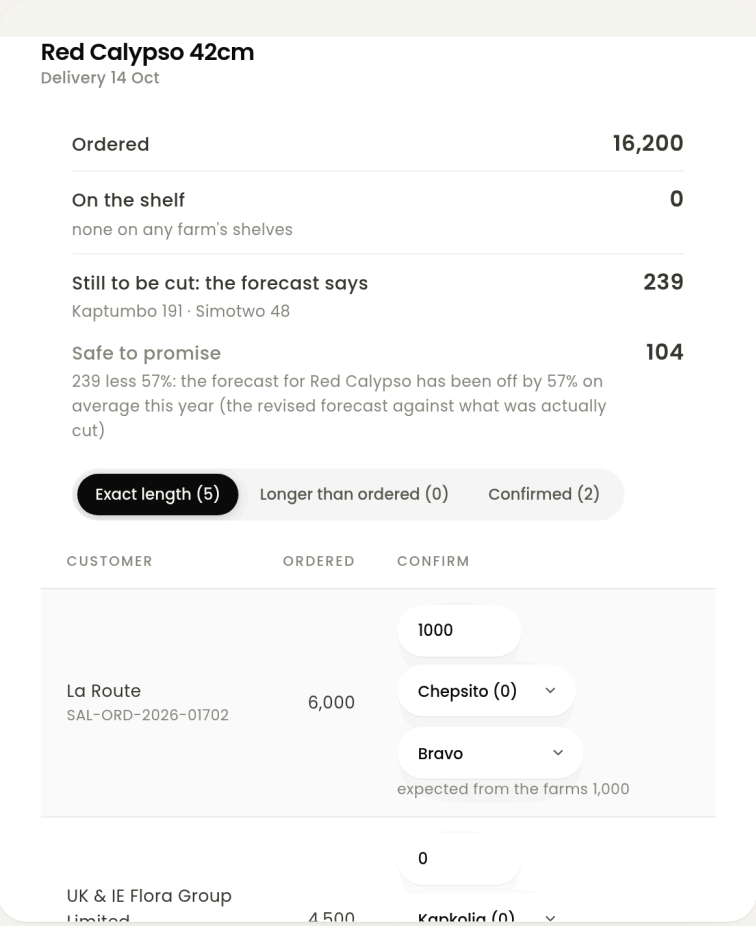
The three tabs

Exact length: orders for this length.
Longer than ordered: orders for a shorter length that this stock could fill.
Confirmed: everyone holding stock here, with Unconfirm and Swap or split.

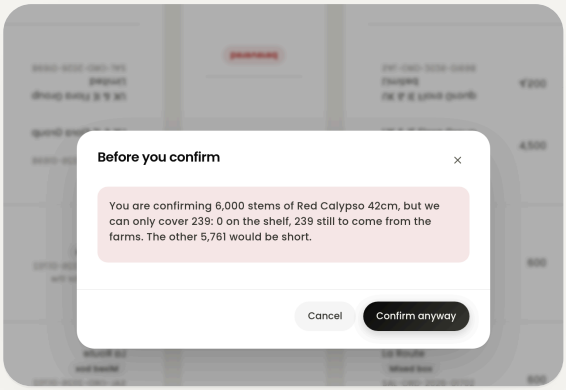
Built to stay readable. No progress bars, no nested tables, three colours only. The stems box stops at what the order allows as you type, and a loader shows the moment a variety is opened.

CONFIRMING

Say what is wrong, then let the planner decide



Detail for one variety: everything that exists, both forecast numbers and why, then a stems box, a From farm and a Team for every customer.



The heads-up popup. It states the facts in one sentence; Confirm anyway goes ahead.

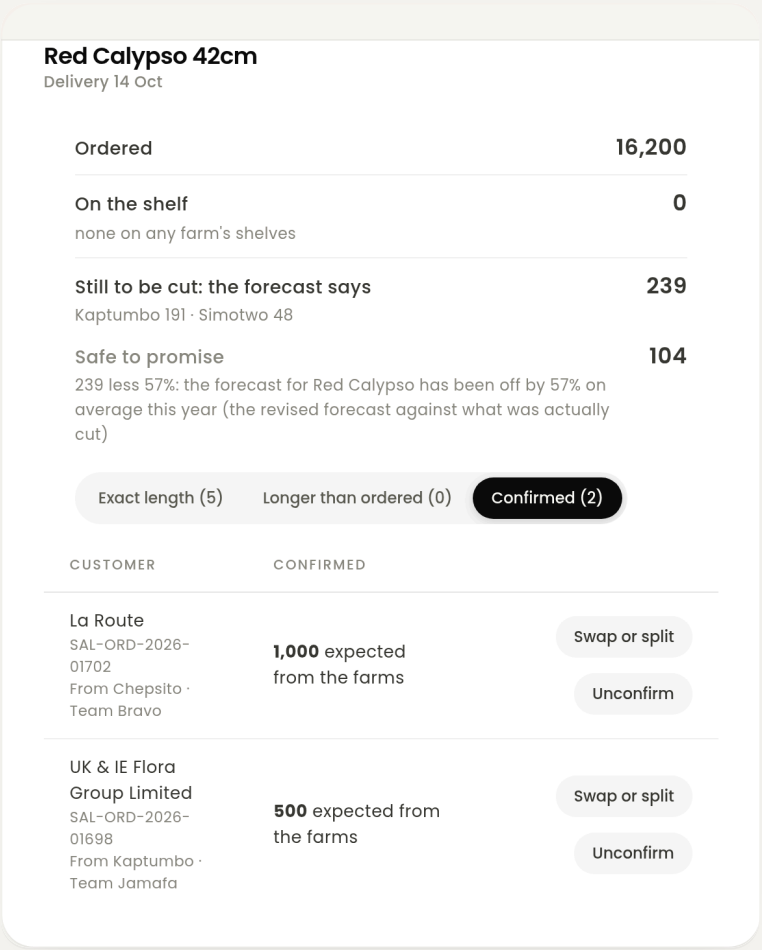
Heads-ups, in plain words

WHEN	WHAT THE PLANNER SEES	COLOUR
Longer than ordered	"If you send 62cm here, Xpol will be short by 1,200 stems. Send 52cm instead?" Quiet when the stock is getting old or nobody else needs it.	red amber
Another customer needs this variety	Names the strict customer who would be short and what this customer also accepts, with a link to the order.	amber
A mixed box cannot be completed	"This box needs Athena 52cm too, and we are short of it by 200 stems."	red
More than we have / are sure of	"...we can only cover 239: 0 on the shelf, 239 still to come. The other 5,761 would be short."	red amber

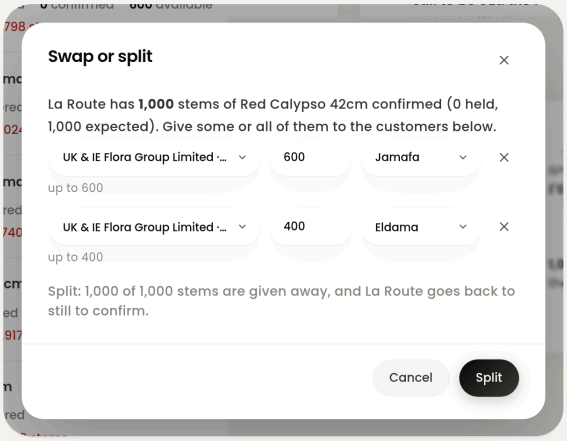
Heads-ups never block. They use each customer's spec (strict means one approved variety or none), the stock for the day and the forecast; every age comes from Production Settings.

AFTER CONFIRMING

See what is confirmed, take it out, swap it or split it



The Confirmed tab: who holds stock here, from which farm, for which team, with Swap or split and Unconfirm.



Giving 1,000 stems to two customers at once.

Swap or split

When stock is confirmed for one customer and another is short, the first customer's row has **Swap or split**. Give everything to one customer and it is a swap; give part, or give to several customers at once, and it is a split.

- What you type is never changed silently; if a customer cannot take it you are told why.
- Stock held on the shelf moves first, then what was confirmed from the expected harvest.
- Both customers are confirmed again, so holds, teams and farms stay consistent.
- A mixed box can receive stems but cannot give them away.

Unconfirm

Takes a whole line out of the plan. What it held goes back to the shelf for everyone, and what was confirmed from the expected harvest is dropped. Setting a line's stems to 0 does the same.

Release happens by itself too

- when allocation uses the stock, and when a line is fully allocated;
- when an order is cancelled or a line changes;
- when the delivery date passes;
- a held bucket that has left the shelf is marked missing and flagged.

UNDER THE HOOD

Records, settings and the link to allocation

Allocation Plan

One per delivery date and sales farm. Closed automatically the day after delivery.

Allocation Confirmation

One per line and source (Real or Virtual): the stems, the stock farm, the team, ordered against planned variety and length, and what has been allocated.

Bucket Reservation

One row per held bucket, indexed by bucket and status. The only table allocation needs to ask "is this bucket held?".

How allocation is linked

Read	The shared stock read subtracts held stems; allocation hides buckets held for another line and refuses them with "held for another order in the plan".
Use	When allocation writes a bucket allocation it consumes the hold for that line.
Release	A fully allocated line lets go of the rest; cancel, change and a passed date release too.
Mirror	Each confirmation is mirrored to the sales order's Confirmed Stems table (farm and stems, real plus expected), which allocation already reads to choose lines and cap stems.
Team	`teams_for_lines` returns the {order line: team} map allocation already accepts, so autofill is one call.

Production Settings

Nothing about ages is written in the code. These existing settings drive the rules: `amber_time`, `discard_age`, `max_shelving_age` and each farm's maximum allocation age and cooling hours.

A new **Allocation Planning** tab holds four plain-labelled values:

- Days from cutting to delivery
- Days ahead people can plan
- Check for new stock every (minutes)
- Tell me when old stock is at least (stems)

The forecast margin is not a setting: it is the forecast's own error.

What changed in the code

NEW	CHANGED
`allocation_plan.py` (holds, release, teams, conversion, mirror) `api/v2/allocation_playground.py` (board, detail, confirm, move) `api/v2/allocation_guards.py` (heads-ups, use first) `api/v2/core/received.py` (received and harvested stock) three records and the planning page, plus the spec	`availability.py` and `core/stock.py` (held stems) `sales_allocation.py` (read, refuse, consume, release) `hooks.py` and `install.py` (jobs, events, settings) the Cold Room page's shelf popups (Full, Half full, Empty)

READY TO ROLL OUT

Checked, with the next steps named

ALLOCATION PLANNING TESTS
33
holds, release, teams, farms, swap and split, forecast

HEADS-UP TESTS
19
every rule, with exact messages

EXISTING SUITES KEPT GREEN
23
old planning page and stock truth

Before you deploy

- Run `bench migrate` : it creates the three records, the Production Settings fields and the indexes, and removes the old forecast-margin field.
- Check the new **Allocation Planning** tab in Production Settings (defaults: 1 day, 3 days, 15 minutes, 500 stems).
- Set `amber_time` explicitly: it is stored as 0 and the allocator quietly treats that as 3 days.
- Check which farms are marked as a sales shelf: they are the choices in the Sales farm filter.
- On a site with a very large Stock Entry table the first open of a day can take a few seconds; after that it reloads in a fraction of a second.

Choices made, so you can change them

- **Another variety is advice, not an action.** The allocator allocates the variety on the order line; no pick row in three weeks used another. The heads-up names who is short and links to the order so sales can change it there.
- **Going ahead needs no reason.** The planner decides.
- **Mixed boxes** are ordinary lines with one shared team; they can receive stems, not give them.
- **Forecast** stays at location level, not per farm.

Not built yet: phase 3

Allocation side
Warn or block allocators who go against the plan, and autofill the team from the plan.

Scorecards
Plan against actual per planner: adherence, downgrade cost, discards avoided, on time in full.

Stock trust
A ghost-bucket rate per shelf, so a shelf count can be trusted by how often it is right.

Honest limits. The development site's stock is weeks old, so the farm-by-farm shelf numbers were checked by tests on real data rather than seen on screen. Offline orders are not tagged yet and count as demand. Nothing in this build has been committed.